

Project Initialization and Planning Phase

Date	29 june 2026
Team ID	
Project Title	Opticrop-smart Agricultural Production Optimization Engine"
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	the primary objective is to revolutionize agricultural production by implementing an advanced machine learning engine (OptiCrop), ensuring fast and accurate data-driven crop recommendations
Scope	The project comprehensively assesses soil and climate factors (N, P, K, pH, rainfall, temperature) to provide intelligent recommendations for farmers and actionable insights for policymakers
Problem Statement	
Description	Addressing the reliance on guesswork and generalized advice in farming, which adversely affects optimal crop selection, resource efficiency, and overall agricultural yields.

Impact	Solving these issues will result in improved operational efficiency for farmers, maximized yields, reduced resource waste, and sustainable agricultural practices.
Proposed Solution	
Approach	Environmental assessment for crop suitability based on precise soil/climate data..
Key Features	- Implementation of a machine learning-based credit assessment model.



	- Real-time decision-making for quicker loan approvals. - Continuous learning to adapt to evolving financial landscapes.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	intel Core i3 or above	T4 GPU
Memory	RAM	4 GB
Storage	Disk space for data,model and logs	10 gb free space
Software		

Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy, matplotlib, seaborn
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv